# Predicting Traffic Accident Severity in Brazil: A Multiclass Machine Learning Approach with Imbalance Handling

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## Contents

# 1 Background and Introduction

## 1.1 Research Questions

This study addresses three research questions:

1. RQ1: How well do standard multiclass classification models predict traffic accident severity?

2. RQ2: How does handling class imbalance affect the performance of these models, and how do the main methods of class imbalance handling compare?

3. RQ3: Which features are most important in predicting traffic accident severity?

## 1.2 Report Structure

# 2 Literature Review

## 2.1 Traffic Accident Severity Classification

When it comes to predicting the severity of an accident, it is important to consider which methods to use. Statistical models, like logit, probit, ordered probit, and multinomial logit, were used before, but recent research papers and tests discovered that machine learning models can outperform them on predictive accuracy. Machine learning models are able to handle non-linear interactions and high-dimensional categorical features, which makes them more ideal for predictions than statistical models.

An early study by Zhang et al. (2018) primarily compared two statistical models for crash severity prediction with four machine learning models. Although all machine learning models achieved better results, they suffered from overfitting. This did not occur with the statistical models, but their results were insufficient. Zhang et al. (2018) concluded that higher accuracy doesn't mean trustworthy insights. However, machine learning models can disagree with each other on variable importance and provide further discussion.

Other studies like Lee et al. (2021) tried to predict the severity of crashes and identify possible risk factors (human, vehicle, road, environmental, etc.) while using only a single CART decision tree. Data from the years 2004 - 2017 for around 170,000 US city crashes were used to predict the final four severity levels. While achieving a prominent 65.78% accuracy, the overall result led to a finding that the model failed to predict any fatal injuries.

Labib et al. (2019) were able to achieve 80% accuracy on four-class severity. However, the dataset consists primarily of data related to fatal accidents, which means that the main results will pertain more to the prediction of the majority class than to performance on the minority class.

Two recent studies targeted the Brazilian PRF dataset and applied machine learning models to predict accident severity, but their limitations motivate the present thesis. Scholz et al. (2024) performed binary predictions on five classifiers, such as Decision Tree, Random Forest, KNN, Naive Bayes, and MLP, on data from the year 2017 to 2022 on the Brazilian PRF dataset, enhanced by including vehicle prices as well, while stating the limitation of not performing any imbalance technique. Meanwhile, Gabardo et al. (2026) applied the imbalance technique, such as SMOTE oversampling, for three classifiers (RF, KNN, MLP), but its limitation lies in the fact that it focused on only one region (southern Brazil) and was used only for binary classification.

Outside the Brazilian context, Ahmed et al. (2023) demonstrated that meaningful metrics are possible to achieve while focusing on using techniques such as SMOTE oversampling, ensemble methods and SHAP for explainability, which leads to a proper understanding of the results. This can be considered a methodological stepping stone for this study, which aims to predict severity across multiple classes using SMOTE oversampling, focusing on the entire country rather than just a specific region.

### 2.2 Class Imbalance Methods

Studies to this point confirm that, when it comes to predicting performance, it is necessary to use appropriate techniques to address class imbalance. The approaches to handle imbalanced data are (Narwane & Sawarkar 2020):

- Data-level - modify the training set (over/under-sampling)
- Algorithm-level - modify the classifier itself
- Cost-sensitive - penalise minority-class errors more heavily
- Ensemble-based - combine multiple weak learners with imbalance-aware schemes

The foundational data-level method SMOTE can create synthetic examples rather than duplicating minority-class instances. This leads the model to a better general understanding of decisions, rather than simply memorising recurring events. SMOTE is a technique to use for imbalanced data to identify fraud detection, medical diagnosis, and crash severity prediction (Chawla et al. 2002).

Although SMOTE is an important technique, it treats all minority classes equally, even though some of them are easier to learn. However, ADASYN is able to generate more synthetic samples for minority classes, which are harder to learn. A limitation of using ADASYN is reduced precision, but in return, recall is increased (He et al. 2008).

To handle noise and class overlap, the hybrid methods such as SMOTE- Tomek and SMOTE-ENN exist. This cleans up ambiguous or noisy samples and enhances performance (Narwane & Sawarkar 2020).

Tanha et al. (2020) performed a benchmark on 19 multi-class datasets to compare methods on imbalanced data. It was discovered that SMOTEBoost consistently outperformed RUSBoost across the benchmark and native multi-class boosters, such as CatBoost and LightGBM, outperformed binary-decomposition OVO approaches.

This confirms the methodological approach of this dissertation, which favors the SMOTE method over the RUS method and highlights the arguments for choosing multiclass classification over binary classification.

### 2.3 Multi-class vs Binary Classification

Previous studies on the severity of traffic accidents have focused on the use of a binary severity classification (Scholz et al. 2024, Gabardo et al. 2026). Although predicting accurate results is essential, binary classification loses important information that could aid in emergency response, resource planning, and policy-making, enabling an appropriate response based on multiple classes, which is more suitable for real-life world scenarios. The difficulty with multi-classification problems comes with handling imbalanced data. Minority classes are more difficult to handle, multi-class imbalance receives less attention from the model and receives less research attention than binary classification (Narwane & Sawarkar 2020). Despite the increased difficulty of performing a multi-class classification model, there is evidence that native multi-class boosting outperforms binary-decomposition (OvO/OvR) approaches on imbalanced data (Tanha et al. 2020). There is a visible natural order in terms of severity. The previous empirical work discovered that using a nominal multi-class model outperformed ordinal models, which leads to the conclusion of using nominal models, even though the data are ordinal (Zhang et al. 2018). Based on these findings, a methodological approach is being developed that focuses on the three-category nominal problem, with the challenge of dealing with data imbalance being addressed.

### 2.4 Explainability and Feature Importance

Simple explainability is important to offer to stakeholders. It is necessary to be able to understand the model prediction. Models with higher accuracy and black-box systems are less valuable than explainable models with less accuracy. Therefore, to be able to assign value to each feature for each individual prediction, SHAP is going to be used. LIME and raw feature importance do not properly guarantee satisfaction of each desirable property, such as local accuracy, missingness, and consistency (Lundberg & Lee 2017). The SHAP model can provide both local explanations and global feature rankings, which is beneficial for answering research question 3. Ahmed et al. (2023) and Scholz et al. (2024) used SHAP to find the most appropriate predictor in their studies, which also supports the decision of using it on tree-based models. Therefore, this leads to the conclusion that SHAP is capable of answering research question 3.

### 2.5 Tree-based Models on Tabular Data

Deep learning or neural network models are becoming a standard approach in many fields. Shwartz-Ziv & Armon (2022) was capable of proving that machine learning algorithms for structured data, such as XGBoost, are capable of dominating 8 out of 11 datasets against four "deep tabular" models (TabNet, NODE, DNF-Net, 1D-CNN). Each deep learning model was capable of performing well on its individual paper, but was unable to adapt to other possibilities. Grinsztajn et al. (2022) findings led to the conclusion that tree-based models are the most appropriate to use on medium-sized datasets based on 45 benchmark datasets testing. This led to the decision to use Random Forest, XGBoost, LightGBM, and CatBoost as the main models with logistic regression as the baseline model for performance evaluation, thereby achieving faster results.

## 2.6 Gaps and Contribution

This literature review addresses three gaps that this dissertation intends to fill.

- Multi-class on Brazilian PRF data

  Previous empirical studies have focused on identifying binary classifications. However, they did not address severity as a problem for multidimensional classification, but merely mentioned it as a topic for future research (Scholz et al. 2024, Gabardo et al. 2026).

- Systematic imbalance-method comparison

  Prior work tends to ignore, acknowledge or only apply one method of handling imbalance data (Ahmed et al. 2023, Gabardo et al. 2026, Lee et al. 2021, Labib et al. 2019). It lacks comparison of using multiple methods across three classifiers, which this dissertation intends to fill (Tanha et al. 2020, Narwane & Sawarkar 2020).

- Modern explainability on PRF data

  This paper focuses on the TreeSHAP algorithm, a specific variant of the SHAP algorithm that is capable of achieving the best performance for multi-class models on the PRF national dataset, which other studies did not elaborate on or investigate (Scholz et al. 2024, Gabardo et al. 2026).

By answering the research questions, this paper addresses the gaps that the literature review established.

# 3 Exploration of Data and Methods

## 3.1 Data Collection

## 3.2 Ethical Approval

## 3.3 Data Description

## 3.4 Data Cleaning

## 3.5 Feature Engineering

## 3.6 Exploratory Data Analysis

### 3.6.1 Target Variable Distribution

### 3.6.2 Geographic Patterns

### 3.6.3 Temporal Patterns

## 3.7 Artefact Design

# 4 Proposed Future Process

# References


Ahmed, S., Hossain, M. A., Ray, S. K., Bhuiyan, M. M. I. & Sabuj, S. R. (2023), 'A study on road accident prediction and contributing factors using explainable machine learning models: Analysis and performance', *Transportation Research Interdisciplinary Perspectives* **19**, 100814.

Chawla, N. V., Bowyer, K. W., Hall, L. O. & Kegelmeyer, W. P. (2002), 'Smote: Synthetic minority over-sampling technique', *Journal of Artificial Intelligence Research* **16**, 321–357.

Gabardo, A. M. P., Schünemann, G. A. A., Jaskowiak, P. A., Moreira, B. G. & Pfitscher, R. J. (2026), Highway to... determining fatal outcomes in traffic accidents based on police reports, *in* 'BRACIS 2025', pp. 230–244.

Grinsztajn, L., Oyallon, E. & Varoquaux, G. (2022), Why do tree-based models still outperform deep learning on tabular data?, *in* 'Advances in Neural Information Processing Systems 35 (Datasets and Benchmarks Track)'.
**URL:** *https://arxiv.org/abs/2207.08815*

He, H., Bai, Y., Garcia, E. A. & Li, S. (2008), ADASYN: Adaptive synthetic sampling approach for imbalanced learning, *in* '2008 IEEE International Joint Conference on Neural Networks (IEEE World Congress on Computational Intelligence)', IEEE, Hong Kong, pp. 1322–1328.

Labib, M. F., Rifat, A. S., Hossain, M. M., Das, A. K. & Nawrine, F. (2019), Road accident analysis and prediction of accident severity by using machine learning in Bangladesh, *in* '2019 7th International Conference on Smart Computing & Communications (ICSCC)', IEEE, Sarawak, Malaysia, pp. 1–5.

Lee, A. S. H., Yap, L. S., Chua, H. N., Low, Y. C. & Ismail, M. A. (2021), 'A data mining approach to analyse crash injury severity level', *Journal of Engineering Science and Technology* pp. 1–14. Special Issue on ACSAT 2021.

Lundberg, S. M. & Lee, S.-I. (2017), A unified approach to interpreting model predictions, *in* 'Advances in Neural Information Processing Systems', Vol. 30, Curran Associates, Inc., Long Beach, CA, USA, pp. 4765–4774.

Narwane, S. V. & Sawarkar, S. D. (2020), 'Machine learning and class imbalance: A literature survey', *Industrial Engineering Journal* . Literature survey covering binary, multi-class, and rare-events class imbalance approaches.

Scholz, J. C. W., Pinheiro, Y. P. & Silva, T. H. (2024), 'Predicting highway accident severity in brazil: Environmental factors and vehicle features', pp. 740–747.

Shwartz-Ziv, R. & Armon, A. (2022), 'Tabular data: Deep learning is not all you need', *Information Fusion* **81**, 84–90.

Tanha, J., Abdi, Y., Samadi, N., Razzaghi, N. & Asadpour, M. (2020), 'Boosting methods for multi-class imbalanced data classification: An experimental review', *Journal of Big Data* **7**(1), 1–47.

Zhang, J., Li, Z., Pu, Z. & Xu, C. (2018), 'Comparing prediction performance for crash injury severity among various machine learning and statistical methods', *IEEE Access* **6**, 60079–60087.

## A Ethical Approval Documentation

## B Use of Generative AI